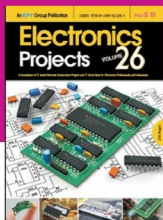


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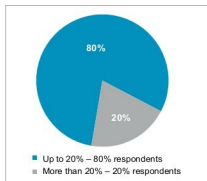


Fig. 2: Expected growth rate of the SMT manufacturing equipment market in India

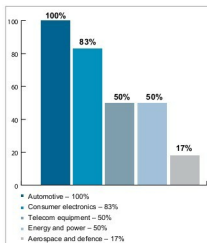


Fig. 3: Survey participants' rankings of the major demand generating application verticals

rooted entrepreneurship spirit, and domestic demand in consumer electronics, telecom equipment, automotive, aeronautics and defence and medical electronics verticals will fuel the growth. Moreover, high-throughput and high-efficiency manufacturing initiatives, based on Industry 4.0 and Smart Factory, will also spur this pace.

The total market can be segmented based on the type of SMT equipment and further sub-segmented based on the technology used (Fig. 1).

Eighty per cent of the survey participants forecast that the Indian market for SMT equipment used in the electronics manufacturing industry will grow at a rate of up to 20 per cent per year (Fig. 2). However, most of them also mentioned sudden dip in the market over the last

six months due to demonetisation and GST related issues.

Market drivers

EMS providers need an increasing number of SMT equipment product lines in order to meet the customer demand across different industry verticals. In addition, many original design manufacturers (ODMs) have also started manufacturing, rather than just designing equipment, in order to overcome the increasing competition from EMS providers. This will further drive the growth of the SMT equipment market.

Adoption of SMT assembly lines will be bolstered further by increasing demand for higher-frequency, highly integrated and smaller devices in consumer electronics, automotive, industrial automation, process control, telecom, Internet of Things (IoT), computing, storage, aerospace and defence markets (see Fig. 3).

The demand for SMT devices in the automotive segment has increased due to their ability to meet the needs of critical systems with zero defects. The number of electronic components used in an electric vehicle (EV) is more than in any conventional vehicle, as the electric vehicle has an electric motor, battery and other complex connections to ensure optimal utilisation of power. Owing to the critical role of electronic components in electric vehicles, surface-mount devices are used as these are safe and lighter, occupy less space and last over a longer period of time.

The consumer electronics segment is transitioning from manual, analogue systems to digitised and automated control systems. With the increase in usage of smart portable devices and wearables, physical constraints on the PCB size are manifold. This could give first-mover advantage to manufacturers who invest in new SMT equipment that support miniature-sized PCBs or components. Feeder technology,